



INDIAN INSTITUTE OF TECHNOLOGY MADRAS

B

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Name **PUNEET**

Total No. of Pages **1**

Quiz I ☐ Quiz II/ Mid-Sem ☐ End-Semester ☐ Make-up ☐ Date : **_____**

Semester & Degree : **_____** Course No. **_____** Part : **_____**

Question No.	1	2	3	4	5	6	7	8	9	10
Marks	2	1.5	2.5	1.5						
11	12	13	14	15	16	17	18	19	20	Total
										7.5

Answer on both sides of the paper including the space below

1.

Researcher

→ name
→ empID
→ base-salary
→ able to calc. total salary

TA → basic info + salary

extra ← (no. of hours + hourly allowance)

Professor → basic info + salary

extra ← (research allowance)

(a) 3 classes
will be -

(i) Researcher

Attributes → name, empID,
salary

Method → total_salary()
init()

(ii) TeachingAssistant

Attributes → name, salary,
empID, no-of-
teaching-hours, hourly-
allowance

Methods → _init_()
total_salary()

(iii) Professor

Attributes → name, salary,
empID,
research_allowance

Methods → _init_()
total_salary()

(b) What we can do is make the Researcher class, then class Teaching Assistant &

Professor can just use the concept on inheritance. They can inherit basic attributes like name, empID, salary and methods total-salary() from the Researcher class and just add few additional information.

2. Projectile, Advanced Projectile
(a) Projectile

(b) Attributes → initial position, speed, launch angle.

1.5

Methods → $\pm$ init - calculate position (t), calculate velocity (t), calculate time (when projectile hits ground)

once more particle flows

Advanced Projectile → initial position, Attributes → speed, launch angle.

Methods → ~~calc~~ init
calc. position (t)
calc. velocity (t)
calc. time (when
projectile hits ground)
get results (t_1, t_2)

↓
in this we can calculate
positions for t_1 & t_2 .

Then we can use any
method to find out the
area under the curve.

Also, we can calculate
few other things like
average velocity during this
time interval.

3. (a) We need ~~one~~ classes to
represent

→ Matrices

→ Row Operations

→ Solver

2.5

(b) Matrices

→ Attributes → init ,

→ Methods → init ,

str

Row operations $\rightarrow$ ~~init~~,
Attributes $\rightarrow$ row, index

Methods $\rightarrow$ - init - , - str -
 $\rightarrow$ extract row from
matrices (using index),
row exchange in
the matrices (using index),
row subtraction in
the matrices (using index)

Solver $\rightarrow$ matrix, vector

Methods $\rightarrow$ - init - , - str - ,
augmentation(), call into row operations
methods inherit from
there.
solve (matrix, vector)

(augmentation) form $\downarrow$
 $Ax = b \Rightarrow [A | b]$
then use row operation
method to get gaussian
decomposition to finally
get the answer

or simply

$x = \text{np.linalg.solve}(A, b)$
gives you results

(c) so as mentioned my plan would to make solver be the child of row operations class. Then I can use

$Ax = b$ to form $[A | b]$ matrix and then using row operations, I can reduce to Gaussian elimination method. And solver for x .

4. (a) $v \times 4 = (8, 6)'$ ✓

(b) $4 \times v =$ it should not work ✓

(c) As far as my understanding goes, I think the problem with this class is. how the input is being taken (self, other) The self has 2 things self.x and self.y. But other is not like that. We need a way such that even if we don't specify the correct position. I should be able $4 \times v$ & $v \times 4$ as the same thing. ✓

We can make other as vector where value 4 gets assigned to both other. $x=4$ and other. $y=4$

In this way, we can make sure
 $u \times v$ & $v \times u$ results in the same
thing.

